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Bwrdd Iechyd Prifysgol
Betsi Cadwaladr
University Health Board

ANAESTHESIA, CRITICAL CARE & PAIN MANAGEMENT DIRECTORATE

OBSTETRIC ANAESTHESIA WRITTEN CONTROL DOCUMENT

GENERAL ANAESTHESIA FOR CAESAREAN SECTION

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1. Introduction and Purpose

General anaesthesia for Caesarean Section carries an increased risk of morbidity and mortality in the parturient. Risks include a 10 fold increase in both failed intubation (1:300) & accidental awareness under general anaesthesia (1:670), along with regurgitation and aspiration of gastric contents. As a result of these risks, use of general anaesthesia for Caesarean Section should be restricted to a limited number of situations (see below).

2. Scope

This guideline is intended as a reference for trainee anaesthetists seeking advice regarding the administration of general anaesthesia for Caesarean Section.

3. Points to consider

3.1 Patient Preparation

- Full preoperative assessment of the patient **(ESPECIALLY AIRWAY)**
- Confirm indication for GA :
    i. Maternal request (despite explanation of GA risks)
    ii. Category 1 Caesarean Section (no time for administration of regional anaesthesia)
    iii. Contraindication to regional anaesthesia :
        a) Coagulopathy (INR >1.4 or platelet count < 80) or full anticoagulation therapy
        b) Sepsis (local or systemic)
        c) Hypovolaemia (eg. antepartum or postpartum haemorrhage)
- Failed attempt at regional anaesthesia, or intraoperative failure
- Additional complex surgery planned at same time as Caesarean Section (where regional anaesthesia not suitable)

- Discussion with the patient of :

    a) GA procedure

    b) Risks of a GA :

        i. Failed intubation (1:300)

        ii. Regurgitation & aspiration of gastric contents

        iii. Accidental awareness under general anaesthesia (1:670)

        iv. Postoperative nausea and vomiting

- Obtain verbal consent for general anaesthesia and document this in the patient's casenotes

- Discussion with the patient regarding postoperative analgesia.

- For Caesarean Sections under general anaesthesia the following postoperative analgesia regime is recommended (if there are no contraindications to these drugs) :

    i. Paracetamol 1G PO 6 hourly regular

    ii. Ibuprofen 600mg PO 6 hourly regular

    iii. Morphine PCA (with ondansetron 4mg im/iv/po 6 hourly regular)

    iv. TAP block (20 - 30ml 0.25% levobupivacaine injected each side and administered either by the surgeon intraoperatively or under USS guidance by anaesthetist immediately following surgery)

    (Use of codeine containing analgesics is contraindicated in breastfeeding mothers (MHRA Guidance))

- For elective Caesarean Sections, patients should be fasted for 6 hours but allowed sips of water until theatre (max 30ml/hour for the last 2 hours prior to theatre)

- Antacid prescription : Omeprazole 40mg at 22:00 (previous day), Omeprazole 40mg at 07:00 (on day of surgery) and 30ml 0.3M Sodium Citrate immediately prior to attending theatre

3.2 Equipment and Staff

- Draw up drugs for GA
- Ensure metaraminol, ephedrine, glycopyrrolate and atropine are at hand
- Check all anaesthetic equipment
- Videoscope should be positioned, switched on, checked and ready for use
- Use additional Oxford HELP pillow if BMI > 45
- Attach nasal specs to patient (NODESAT)
- Ensure ODP is present prior to commencing general anaesthesia
- **The resident anaesthetist covering obstetrics should request consultant help if a general anaesthetic needs to be given in a patient with a predicted or known difficult airway**
- **The resident anaesthetist covering obstetrics, in the incidence where there has been several failed attempts at spinal anaesthesia, and there is no immediate threat to maternal or fetal life, should request consultant (or other senior anaesthetist) help to attempt a spinal anaesthetic before proceeding to a general anaesthetic**

3.3 Technique

- Attach full routine monitoring (ECG, NIBP, Pulse Oximetry)
- Site a **16G** (or 14G) iv cannula
- Connect iv infusion of **Hartmann's solution**
- Place patient in **supine position with 15 degree left lateral tilt**
- Administer iv antibiotic (Cefuroxime 1.5g, and metronidazole 500mg iv or if past history of anaphylaxis to penicillin give Clindamycin 600mg & Gentamycin 120mg iv)
- Rapid sequence induction (cricoid pressure) :

    i. Commence oxygen at 4 litres / min via nasal specs to optimise preoxygenation and extend apnoea time

    ii. Preoxygenate the patient **using 100% oxygen** via circle with neat fitting facemask until ET O2 > 0.9

    iii. Ensure **Pre-Induction Checklist for Obstetric GA** is completed by ODP during preoxygenation and prior to commencing general anaesthesia

iv. Induce anaesthesia when surgical, paediatric and midwifery staff are ready

v. Thiopentone (5 - 8 mg / Kg) (or alternatively Propofol (2.5 – 2.8 mg / Kg))

vi. Suxamethonium (1.5mg / Kg)
(Use rocuronium (1.2mg / Kg) if airway assessment predicts difficulty with intubation)

vii. Increase oxygen flow via nasal specs to 10 litres / min

- **Intubation in obstetrics should <u>always</u> be undertaken using a videoscope**
- Intubate using a size 7mm cuffed oral endotracheal tube (COETT)
- Ask ODP to release cricoid pressure when :

  i. Position of COETT confirmed using capnograph and auscultation of chest

  ii. No air leak detected after cuff inflated

- Secure COETT
- Ventilate with a mixture of 33% Oxygen / 67% Nitrous Oxide + Sevoflurane (maintain a minimum of 1.2 MAC (age adjusted)
- Give Rocuronium (0.4 mg / Kg) as maintanance muscle relaxant
- After delivery of the baby :

  i. Administer 5iu bolus dose of syntocinon

  ii. Administer 10 mg morphine (in 5mg increments)

  iii. **If requested by surgeon** (for atonic uterus) set up infusion of 40iu syntocinon in 500mls 0.9% Saline and run at a rate of 125 mls / hr

- Administer sodium diclofenac 100mg suppository PR at end of procedure (if no contraindications)
- Also at end of procedure ensure a minimum of 3 muscle contractions in response to TOF prior to the administration of the neuromuscular reversal 2.5mg neostigmine + 0.5mg glycopyrrolate, otherwise administer Sugammadex (2 – 4mg/Kg)
- Ensure T.O.F. > 90% prior to extubation
- Perform a TAP block for postoperative analgesia (if not contraindicated or already undertaken by the surgeon)

- Extubate patient sitting up or in full left lateral position when :

    i. Adequate tidal volume and respiratory rate

    ii. Full return of laryngeal reflexes

    iii. Patient obeys commands eg.opens eyes

- Following surgery proceed to the recovery area with the patient. Do not leave until the patient is fully conscious, pain free and has stable vital signs

3.4 Troubleshooting

- If poor view on videolaryngoscopy then :

    i. Adjust and optimise head and neck position (sniffing morning air position)

    ii. Adjust cricoid pressure (ensure it is applied perpendicular to the operating table), consider relaxing or even releasing cricoid pressure if view of the glottis is still poor

    iii. Change the blade on the videoscope from a C blade to a D blade (and use a stylet instead of a bougie to intubate)

- If you have two failed attempts at intubation despite the above manoeuvres then **Call for Help**, manually ventilate with 100% oxygen plus volatile agent (1.2 MAC) and follow the **OAA / DAS Obstetric Failed Intubation Guideline** (copy available in red emergency folder)

- If systolic BP is below 100mmHg (or below 20% baseline) at any time pre delivery then increase rate of iv fluid, give metaraminol boluses (0.5 mg) and consider lightening anaesthesia (but remember that MAC (age adjusted) should always be 1.2 or higher to reduce the risk of awareness)

4. References

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3. NAP5. 5th National Audit Project Report : Accidental awareness under general anaesthesia in the UK & Ireland (2014) RCoA & AAGBI, London